

Firdawss El Haddouchi

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About Me

Data Engineering student skilled in building scalable data pipelines using Spark, Airflow, and Trino. Passionate about transforming ‘messy’ raw data into actionable insights through automated and containerized solutions.

Education

National School of Applied Sciences, First-Year Engineering Degree in Data Engineering – Al Hoceima, MOROCCO 2024 – present

National School of Applied Sciences, Preparatory Classes in Engineering Degree – Al Hoceima, MOROCCO 2022 – 2023

Experience

Software Development Intern (Healthcare IT), Regional Hospital Center of Al Hoceima – Al Hoceima, Morocco July 2025 – Sept 2025

- Participated in the end-to-end development and deployment of a web-based management system for the Radiology Service.
- Contributed to backend and frontend development using PHP, JavaScript, and MySQL, ensuring high data integrity.
- Collaborated with a technical team to analyze requirements and deliver a functional solution deployed on a central server.
- Implemented interactive reporting modules using Chart.js to monitor departmental performance and patient metrics.

Business Intelligence Intern, Terra Nova EBS – Al Hoceima, Morocco July 2025 – Aug 2025

- Contributed to the development of interactive BI dashboards using Power BI to transform business data into actionable insights.
- Participated in data modeling and the creation of advanced analytical measures using DAX.
- Leveraged Python to support data analysis and generate basic predictive insights from raw datasets.
- Collaborated within a team to ensure data accuracy and effective visualization of key business metrics.

Projects

TaaSim: Smart Urban Mobility Platform (Casablanca) Mar 2026 – present

- Designing a real-time data processing architecture to simulate GPS data flows and trip requests for urban mobility.
- Integrating Apache Kafka for high-throughput message ingestion and Apache Flink for stream processing and real-time analytics.
- Utilizing Apache Spark for batch processing of historical mobility patterns to optimize route recommendations.
- Ensuring scalable deployment using Docker containers to manage the distributed ecosystem.

JobIntelligent: AI-Powered Job Analytics Platform Mar 2026 – May 2026

Conceived and implemented an end-to-end Big Data platform for job market analysis and personalized recommendations using a Medallion Architecture.

- Built a robust data pipeline using Apache Airflow to orchestrate daily ingestion from international APIs (Adzuna, Reed,arbeitnow) into a MinIO Data Lake (Bronze/Silver layers).
- Developed a NLP-based Resume Parser using Python to automatically extract technical skills and provide personalized job matches via Fuzzy Matching.
- Designed a modern web interface (SPA) with FastAPI and Glassmorphism UI, featuring secure JWT authentication and real-time job tracking.
- Modeled a Star Schema in PostgreSQL (Gold layer) to power interactive Power BI dashboards for salary trends and geographic market analysis.

Batch Procurement Data Pipeline Dec 2025 – Jan 2026

Built an end-to-end data pipeline to automate supply chain flows using a hybrid storage architecture.

- Managed high-volume daily transactional sales data using Hadoop HDFS and PostgreSQL master data.
- Utilized Trino as a unified query engine to federate distributed queries, significantly reducing data movement.
- Processed diverse data formats (Avro/Parquet) and ensured environment consistency via Docker containerization.

Database Performance & Optimization Study

Nov 2025 – Nov 2025

Analyzed query performance in large-scale Oracle environments using advanced PL/SQL.

- Simulated a high-volume environment with millions of rows to analyze latency patterns.
- Used DBMS_XPLAN to optimize execution plans, reducing response times for complex queries.

Skills

Data Engineering: Apache Spark, Apache Kafka, Apache Flink, Apache Airflow, Trino, Hadoop HDFS, ETL Pipelines, Medallion Architecture

Databases & Analysis: MySQL, PostgreSQL, Oracle SQL, PL/SQL, Pandas, NumPy, Power BI (DAX)

Programming & AI: Python, Java (Spring Boot), Machine Learning, NLP

Tools & DevOps: Docker, Git, GitHub, T-SQL

Languages

Arabic & Riffian: Native

English & French: Intermediate (A2 - actively improving)

Certifications

- **Introduction to Docker** — DataCamp: Gained hands-on experience in containerizing applications and managing images/containers for scalable deployments.
- **Supervised Learning with Scikit-Learn** — DataCamp: Built and evaluated predictive models using regression and classification algorithms for data-driven insights.
- **Understanding Artificial Intelligence** — DataCamp: Mastered foundational AI concepts, ethics, and their practical applications in business and technology.
- **Power BI Online Course** — Udemy: Developed advanced data visualization skills, creating interactive dashboards with complex DAX measures.

Qualities

- Strong analytical thinking and problem-solving skills.
- Excellent teamwork and collaboration abilities.
- Commitment to precision, accuracy, and continuous learning.